

Comprehensive analysis of nasal IgA antibodies induced by intranasal administration of the SARS-CoV-2 spike protein

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Abstract

Intranasal vaccination is an attractive strategy for preventing COVID-19 disease as it stimulates the production of multimeric secretory immunoglobulin A (IgAs), the predominant antibody isotype in the mucosal immune system, at the target site of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) entry. Currently, the evaluation of intranasal vaccine efficacy is based on the measurement of polyclonal antibody titers in nasal lavage fluid. However, how individual multimeric secretory IgA protects the mucosa from SARS-CoV-2 infection remains to be elucidated. To understand the precise contribution and molecular nature of multimeric secretory IgAs induced by intranasal vaccines, we developed 99 monoclonal IgAs from nasal mucosa and 114 monoclonal IgAs or IgGs from nonmucosal tissues of mice that were intranasally immunized with the SARS-CoV-2 spike protein. The nonmucosal IgAs exhibited shared origins and both common and unique somatic mutations with the related nasal IgA clones, indicating that the antigen-specific plasma cells in the nonmucosal tissues originated from B cells stimulated at the nasal mucosa. Comparing the spike protein binding reactivity, angiotensin-converting enzyme-2-blocking and SARS-CoV-2 virus neutralization of monomeric and multimeric IgA pairs recognizing different epitopes showed that even nonneutralizing monomeric IgA, which represents 70% of the nasal IgA repertoire, can protect against SARS-CoV-2 infection when expressed as multimeric secretory IgAs. Our investigation is the first to demonstrate the function of nasal IgAs at the monoclonal level, showing that nasal immunization can provide effective immunity against SARS-CoV-2 by inducing multimeric secretory IgAs at the target site of virus infection.

eLife assessment

This work provides **valuable** insights into mucosal antibody responses against SARS-CoV-2 following intranasal immunization by characterizing a large number of monoclonal antibodies at both mucosal and non-mucosal sites. The evidence supporting the claims is overall **solid**, although the flow cytometric assessment of antibody-expressing cells would benefit from more rigorous controls. The demonstrated in vitro antiviral activity of antibodies characterized provides a rationale for developing mucosal vaccines, especially if confirmed in vivo and benchmarked against antibodies generated following intramuscular vaccination.

Introduction

Immunoglobulin A (IgA) is differentially distributed between the systemic and mucosal immune systems ([Li et al., 2020](#) [DOI](#)). Monomeric IgA (M-IgA) is predominantly present in serum, whereas secretory IgA (S-IgA) is the most prevalent IgA in mucous secretions. S-IgA is composed of two IgA units and one J chain, which are synthesized and assembled in local plasma cells. Secretory components expressed on the basolateral surface of mucosal epithelial cells bind the IgA complex through the J chain and transport the molecule to the apical cell membrane ([Woof and Kerr, 2006](#) [DOI](#)). Recent studies have shown that S-IgA in the nasal mucosa exists not only as dimers but also as multimers (trimers and tetramers) in the human upper respiratory mucosa ([Saito et al., 2019](#) [DOI](#)). S-IgA plays an important role in the protection and homeostatic regulation of the airway, intestinal and vaginal epithelium through a process known as immune exclusion and immunosuppression ([Matsumoto, 2022](#)). Many studies have shown that S-IgA on mucosal surfaces is more effective and more cross-protective than IgG in serum for protection from harmful pathogens ([Okuya et al., 2020b](#) [DOI](#)) ([Asahi et al., 2002](#) [DOI](#)) ([Dhakal et al., 2018](#) [DOI](#)) ([Asahi-Ozaki et al., 2004](#) [DOI](#)).

COVID-19 is a disease caused by infection with severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) that causes mild upper respiratory symptoms in most cases, but some patients develop bilateral pneumonia with acute respiratory distress ([Budinger et al., 2021](#) [DOI](#)). The SARS-CoV-2 spike protein expressed on the virus surface is a multidomain homotrimer protein composed of an S1 domain consisting of an N-terminal domain (NTD) and receptor binding domain (RBD) and an S2 domain that mediates fusion of the virus and host cell membrane. Viral infection is initiated through the interaction between the RBD and the host receptor angiotensin-converting enzyme-2 (ACE2) ([Jackson et al., 2022](#) [DOI](#)). Thus, the spike protein is the main target for current vaccine development because antibodies against this protein can neutralize the infection. Currently approved intramuscular COVID-19 vaccines predominantly induce a systemic immune response by producing IgG in serum before they cause severe tissue damage, resulting in a high degree of efficacy ([Polack et al., 2020](#) [DOI](#)). However, these vaccines mainly induce IgG and M-IgA in serum; they do not induce S-IgA, which coats the upper respiratory tract mucosal surface ([DeFrancesco, 2020](#) [DOI](#)). The emergence of highly infectious SARS-CoV-2 omicron variants may undermine the therapeutic efficacy of vaccines, requiring more effective vaccination to prevent SARS-CoV-2 infection. Since nose epithelial cells are a primary target for SARS-CoV-2, intranasal vaccinations that induce S-IgA in the upper respiratory tract are desirable for protection against the infection and transmission of the virus ([Sunagak et al., 2020](#) [DOI](#)). To date, some intranasal vaccines are under development and have shown a robust mucosal and humoral immune responses in human and animal models ([Houston, 2023](#) [DOI](#)) ([Chavda et al., 2021](#) [DOI](#)) ([Alu et al., 2022](#) [DOI](#)) ([Barrett et al., 2021](#) [DOI](#), [Bricker et al., 2021](#) [DOI](#), [D'Arco et al., 2021](#) [DOI](#), [Kim et al., 2021](#) [DOI](#)) ([Vabret et al., 2020](#) [DOI](#)) ([Ohtsuka et al., 2021](#) [DOI](#)). Considerable efforts have been made to develop mucosal

vaccines against pathogens, and nasal lavage fluids containing polyclonal S-IgAs have been used to evaluate the response to these vaccinations ([Maltseva et al., 2022](#)) ([Gianhecchi et al., 2019](#)) ([Wong et al., 2022](#)) ([Afkhami et al., 2022](#)) ([Sui et al., 2021](#)) ([Azzi et al., 2022](#)). However, it is difficult to obtain information regarding the functions of individual antibodies that comprise the polyclonal anti-viral mucosal repertoire from such broad assays. It has also been shown that intranasal vaccination induces a systemic antibody response, but the origin of these antibodies has not previously been investigated at the molecular level ([Tumpey et al., 2001](#)).

To understand the precise contribution and molecular nature of S-IgA induced by intranasal vaccines with respect to its antiviral function, the development of monoclonal S-IgAs from plasma cells localized in nasal mucosa and studying their function is essential. However, because of the difficulties in developing monoclonal IgA antibodies from nasal mucosal tissue, many studies have used artificial switching of IgG to IgA for recombinant production, and this approach has been used to study the protective properties of IgA against pathogens ([Ejemel et al., 2020](#), [Saito et al., 2019](#)). Such technological limitations have hampered the biochemical and clinical evaluation of intranasal vaccination at the molecular level. Our group has established robust protocols for the isolation of single antigen-specific plasma cells from a variety of animals and the high-throughput production of monoclonal antibodies ([Kurosawa et al., 2012](#)).

Here, using this technology, we developed 213 antigen-specific monoclonal antibodies from plasma cells derived from mice that were intranasally immunized with a stabilized SARS-CoV-2 spike trimer protein (Spike) and demonstrated that intranasal immunization induced nose-related antibodies in the spleen, lung, and blood. Analysis of the properties of monoclonal IgAs recognizing different epitopes revealed that multimerization of M-IgAs could induce neutralizing activity.

Results

Development of anti-Spike monoclonal M-IgAs from plasma cells localized in nasal mucosa

To evaluate the immune response after the intranasal administration of the SARS-CoV-2 spike protein of the Wuhan-Hu-1/D614G strain (Spik_{Wuhan}), nasal lavage fluids and serum were collected from the mice one week after the last immunization. The intranasal administration of Spik_{Wuhan} induced high levels of antigen-specific IgA but not IgG in nasal lavage fluids compared with those of the phosphate-buffered saline (PBS) control (**Fig. 1A**). Elevated levels of antigen-specific IgA and IgG responses were found in serum taken from the immunized mice, which is consistent with a previous study showing that intranasal vaccinations induce both nasal and serum IgA levels ([Sterlin et al., 2021](#)) ([Maltseva et al., 2022](#)). We selected three immunized mice with high nasal IgA titers and isolated antigen-specific plasma cells from the nasal mucosa, spleen, lung and blood. Cells isolated from these tissues were stained with fluorescently labeled anti-CD138, ER-tracker ER-Tracker Blue-White DPX, anti-IgA and S1 subunit of Spik_{Wuhan} (S1), and then antigen-specific plasma cells defined as anti-CD138⁺, ER-tracker^{high}, anti-IgA⁺ and S1⁺ were isolated by fluorescence-activated cell sorting (FACS). Consistent with the pattern of the IgA response observed in nasal lavage fluid, the proportions of antigen-specific IgA⁺ plasma cells in the nasal mucosa of immunized mice were significantly higher than those of control mice (**Fig. 1B** and Supplementary file 1A).

Cognate pairs of immunoglobulin heavy chain variable (VH) and kappa light chain variable genes (VL) were amplified by rapid amplification of 5' cDNA ends PCR from the single-sorted cells. After constructing full-length immunoglobulin heavy and kappa light chain genes, antibodies were expressed by DNA transfection into CHO-S cells. Antigen-specific antibody clones were identified by enzyme-linked immunosorbent assay (ELISA) with immobilized Spik_{Wuhan}. After sequencing of

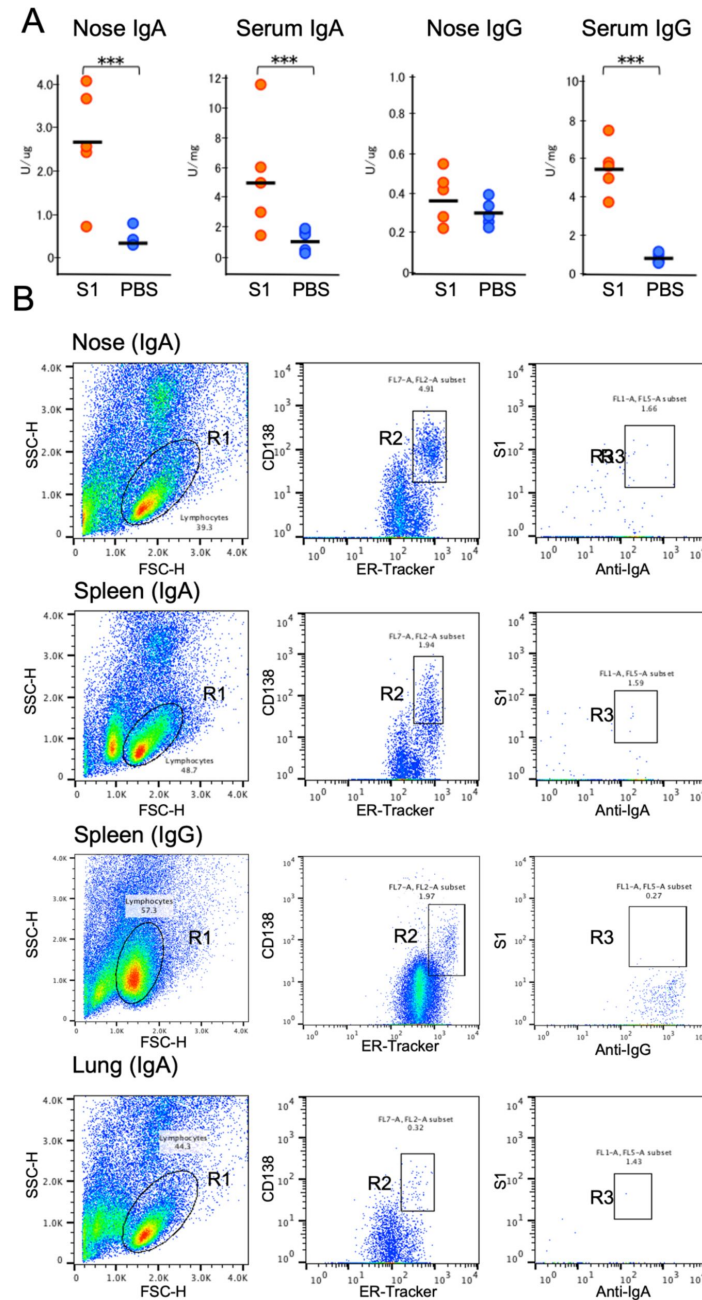


Fig. 1.

Isolation of antigen-specific IgA⁺ plasma cells from mice that were intranasally immunized with SpikWuhan.

(A) Mice were inoculated via the intranasal route with 10 µg of SpikWuhan or PBS with 1 µg CT as an adjuvant. Nasal lavage fluid and serum were collected from the mice at one week after the last immunization, and antibody responses were evaluated using ELISA.

(B) FACS gating strategy for the isolation of S1-specific plasma cells from mice.

Cells from nasal mucosa or spleen were stained with anti-CD138, ER-tracker, anti-mouse IgA, and S1. Plots (I)-(III) represent the sequential gating strategy. (I) FSC vs. SSC with gate R1 representing lymphocytes. (II) The CD138⁺ ER-Tracker^{High} fraction was defined as plasma cells. (R2). (III) The S1-specific plasma cells were defined as IgA⁺, ER-Tracker^{High} and S1⁺ (R3 gate). The numbers indicate the percentages of cells in the gated area. A total of 100,000 events were recorded. Representative data from the No. 1 mouse are shown.

the entire coding region of heavy and light chain genes, the antibodies sharing the same V-(D)-J genes were grouped (Supplementary Table 1). Then, representative antibody clones from each group were analyzed for their binding to the spike protein RBD of Wuhan, Beta, Kappa and Delta variants and an NTD of Spik_{Wuhan}. The antibodies were also analyzed for their ability to block ACE2 binding to RBDs and to neutralize Wuhan pseudotyped virus. Based on their properties, each group of antibodies was arbitrarily categorized into five types. Type 1: anti-RBD, *ACE2 blocking neutralizing antibody*; Type 2: anti-RBD, *ACE2 blocking nonneutralizing antibody*; Type 3: anti-RBD, *non-ACE2 blocking nonneutralizing antibody*; Type 4: anti-NTD, *non-ACE-2 blocking nonneutralizing antibody*; and Type 5: *non-ACE-2 blocking nonneutralizing antibody targeting epitopes other than RBD and NTD*.

Intranasal immunization induces functionally diverse IgA in the nasal mucosa and spleen

We conducted detailed characterization of antibodies obtained from the No. 1 mouse, as many antigen-specific M-IgA clones were obtained. Of the 51 nasal M-IgA clones analyzed, they were classified into 11 groups based on their V-(D)-J usage, with the majority (83%) clonally expanding into four major clusters (G2, G3, G4 and G10) and the remainder (17%) scattered across branches (**Fig. 2A and B** [↗](#)). The antibodies in the G1-G4 clusters were categorized as Type 1. The G1 antibody appeared only once. This clone displayed potent binding to the RBD of Wuhan, Kappa and Delta, and moderately to the Beta variant. It was also able to block ACE2 binding to all RBDs except the Beta variant RBD. The antibodies in G2-4 clusters bound to the RBD of Wuhan and Delta but not to the RBD of Beta and Kappa. They were able to block ACE2 binding to the RBD of Wuhan and Delta but not to the RBD of Beta and Kappa. The antibodies in G5-G9 clusters were categorized as Type 2 and bound to all four RBD variants and uniformly blocked ACE2 binding to the RBDs of all four strains. G10 antibodies were categorized as Type 3 and represented 38% of the nasal IgA repertoire. They bound to the RBDs of all four strains but failed to block ACE2 binding to the RBDs of all four strains. The G11 antibody was categorized as Type 4, was an anti-NTD antibody and did not block ACE2 binding to the RBDs of all four strains.

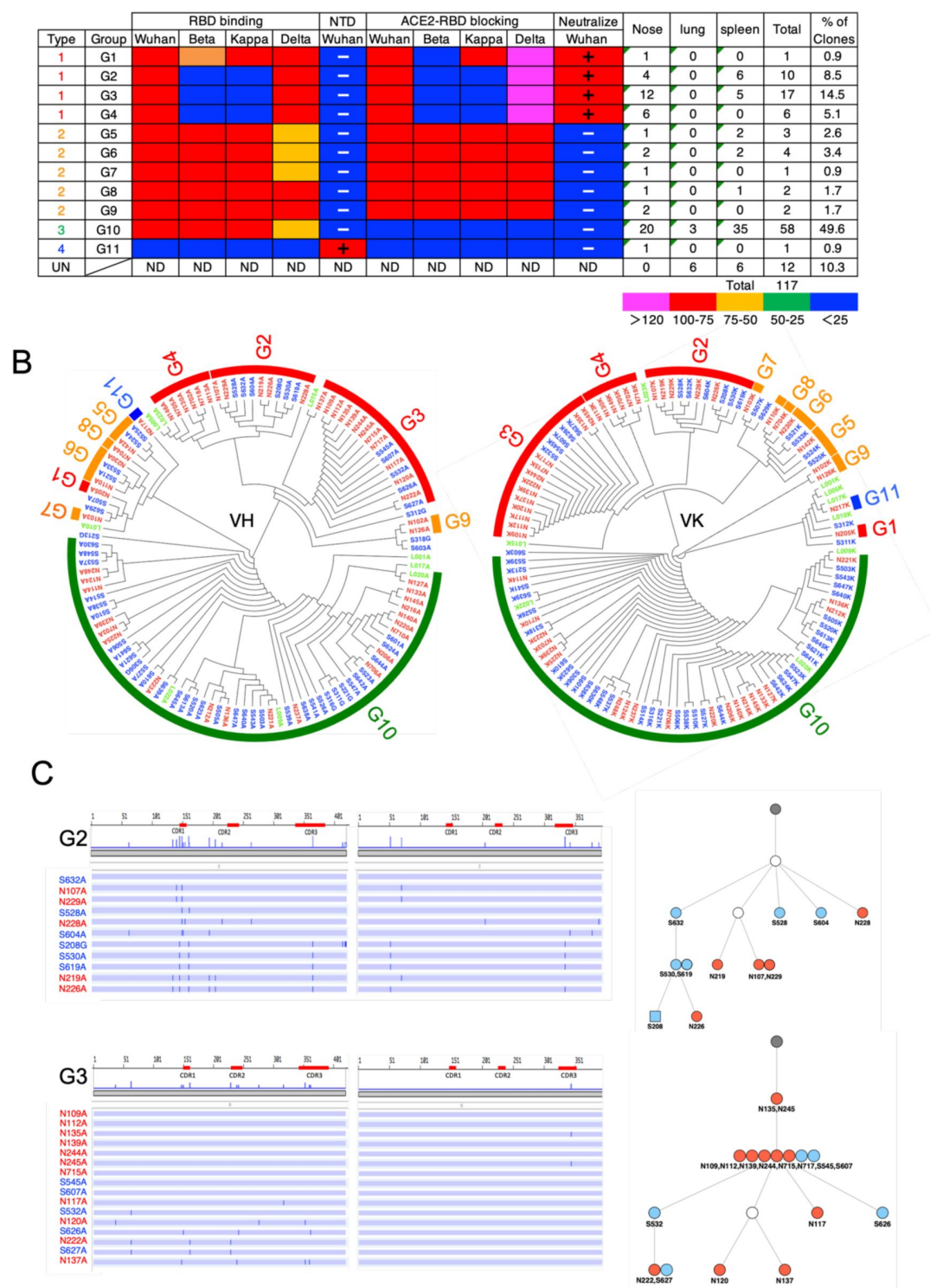


Fig. 2

Intranasal immunization induces functionally diverse antibodies in the nasal mucosa and spleen.

(A) Characterization of S1-specific monoclonal antibodies obtained from No. 1 mouse. The heatmap represents the relative intensity of antibody binding to RBDs and blocking of the RBD-ACE2 interaction. Blue (0–25%), green (25–50), orange (50–75%) and red (>75%). NTD binding was considered positive (+) when the OD at 405 nm was >0.3 after the

background was subtracted. Neutralizing activity was considered positive (+) when the antibody suppressed Wuhan pseudotyped virus infection by 50%. The figure reports values from a single experiment. UN, antibody type not determined. ND, antibody activity not determined.

(B) Maximum-likelihood phylogenetic tree of the VH and VL chains of the S1-specific antibodies.

Different colored fonts indicate antibodies obtained from the nose (red), spleen (blue), and lung (green). Antibody groups are indicated by bands on the outer ring. The color of the band indicates antibody types: Type 1 (red), Type 2 (orange), Type 3 (green), Type 4 (blue) and Type 5 (gray). The antibody group is defined as clones using the same V-(D)-J usage and having an overall sequence identity of at least 95% from the signal peptide to framework 4 (FR4). The prefixes N, S and L in the antibody clone numbers refer to antibodies derived from the nose, spleen and lung, respectively. The suffixes A, G and K in the antibody clone numbers refer to alpha, gamma and kappa chain, respectively.

(C) Nucleotide sequence arraignment of VH and VL genes in the G2 and G3 antibodies from No. 1 mouse. The VH and VL sequences from the beginning of the signal peptide through the end of FR4 are shown as horizontal lines. Nucleotide changes relative to S632A and N109A are depicted as vertical bars across the horizontal lines. Different colored fonts indicate antibodies derived from the nose (red) and spleen (blue). Antibody phylogenetic trees based on VH/VK paired sequences are depicted. Gray circles represent the hypothetical germline configuration. White circles represent hypothetical ancestors. Colors indicate nasal (red) and splenic (blue) antibodies. Circles and squares indicate IgA and IgG, respectively.

Intranasal immunization induces nose-related antibodies in the spleen, lung, and blood

Previous reports suggest that nasopharynx-associated lymphoid tissue preferentially selects high-affinity IgA⁺ B cells, which not only home back to regional mucosa but also migrate into nonmucosal tissues (Shimoda *et al.*, 2001). If this is the case, a fraction of S1-specific plasma cells differentiated from nose-originated B cells may reside in the spleen and produce antibodies. To directly evaluate the cellular origin of the anti-S1 antibodies in nonmucosal tissues, we analyzed the presence of S1-specific plasma cells in the spleen, lung, bone marrow and Peyer's patches. FACS analysis of splenocytes harvested from No. 1 mice showed the presence of S1-specific IgA⁺ plasma cells, but the splenocytes harvested from control mice did not exhibit S1-specific IgA⁺ plasma cells (Fig. 1B and Supplementary file 1B). Single-cell-based immunoglobulin gene cloning resulted in the successful production of 49 S1-specific monoclonal IgAs. DNA sequence analysis of these clones revealed significant clonal overlap between the nose and spleen. (Fig. 2A and B). Clonal overlap was found in the G2-G3, G5, G6 and G10 clusters, in which G10 possessed the most expanded splenic clones, as in the case of nasal IgA (Fig. 2C). Further investigation was carried out to determine whether IgG⁺ plasma cells that express nose-related IgA could be detected in the spleen. Although the number of antigen-specific IgG⁺ plasma cells was limited, eight IgG clones specific for S1 were isolated, among which five were nose-related. When we attempted to isolate antigen-specific IgA⁺ plasma cells from the lung, nine S1-specific IgA clones were obtained, among which three were nose-related. The gut mucosa and bone marrow possessed a large population of IgA⁺ plasma cells. However, we could not detect S1-specific IgA⁺ plasma cells in Peyer's patches and bone marrow (Supplementary file 1B).

Next, we focused on the expanded antibody groups G2, G3 and G10 to analyze the patterns of somatic hypermutation (SHM). We found complex patterns of shared and unique SHMs in antibodies obtained from the nose, spleen and lung. In G2, the VHs of the three splenic clones (S530A, S208G, and S619A) had the same sequence, except for framework 4 of an IgG clone (S208G) and had three shared SHMs with the nasal clones (N219A and N226A). In addition, the VKs of the three splenic clones and N226A had the same sequence. In Group 3, seven nasal clones (N109A, N112A, N135A, N139A, N244A, N245A and N715A) and two splenic clones (S545 and S607) were 100% identical in nucleotide sequence, with N135A and N245 differing from the other clones by

only one nucleotide in their VKs (**Fig. 2C**). Sequence analysis of the most expanded group, Group 10, also demonstrated clonal overlap in the nose, lung and spleen. For example, the VHs of a nasal clone (N221), spleen clones (S647, S503, S640 and S543) and a lung clone (L009) were 100% identical, with S647 differing from the other clones by only one nucleotide in VK (Supplementary file 2). Antibody lineage analysis of these clones suggests that each family of related antibodies originates from a common ancestor gene that was subjected to class switching and SHM. We also developed S1-specific Abs from No. 2 and No. 3 mice and analyzed their sequences (**Fig. 3** and Supplementary file 3). In No. 2 mice, we obtained 29 nasal IgAs, 15 splenic antibodies (13 IgAs and two IgGs), five lung IgAs and six blood IgAs. They were classified into 17 groups based on their V-(D)-J usage, and each group was categorized as either Type 1, 2, 3, 4 or 5 based on antibody properties. (**Fig. 3**). Nose-related clones were found in spleen, lung and blood (G5, 6, and 8). These clones also showed a complex pattern of shared and unique SHM with nasal clones throughout the full length of the VH and VL genes. In mouse No. 3, we cloned 19 nasal IgAs and 22 splenic antibodies (17 IgAs and five IgGs). They were classified into 10 groups, and each group was categorized as either Type 1, 3, 4 or 5. A high degree of clonal overlap between the nose and spleen was found, in which 13 out of 22 splenic Abs were nose-related clones (Supplementary file 3). Analysis of the VH and VL repertoires of the three mice revealed that the expanded clones constituted varying fractions of the antibody repertoire among different mice despite having received the same antigen, and no group of antibodies stood out across mice, suggesting that individual mice had immunologically distinct responses (Supplementary figure 4). In all mice, only a few bound to the NTD, and most of them were RBD-directed. Approximately 30% of M-IgAs categorized as Type 1 showed neutralizing activity (Supplementary figure 4). Taken together, regarding the mutation frequency and the pattern of SHM, it can be assumed that B cells stimulated by nasal challenges were the major precursor of antigen-specific plasma cells in the spleen, lung and blood, which may contribute to antibody production in the lower respiratory tract and systemic circulation.

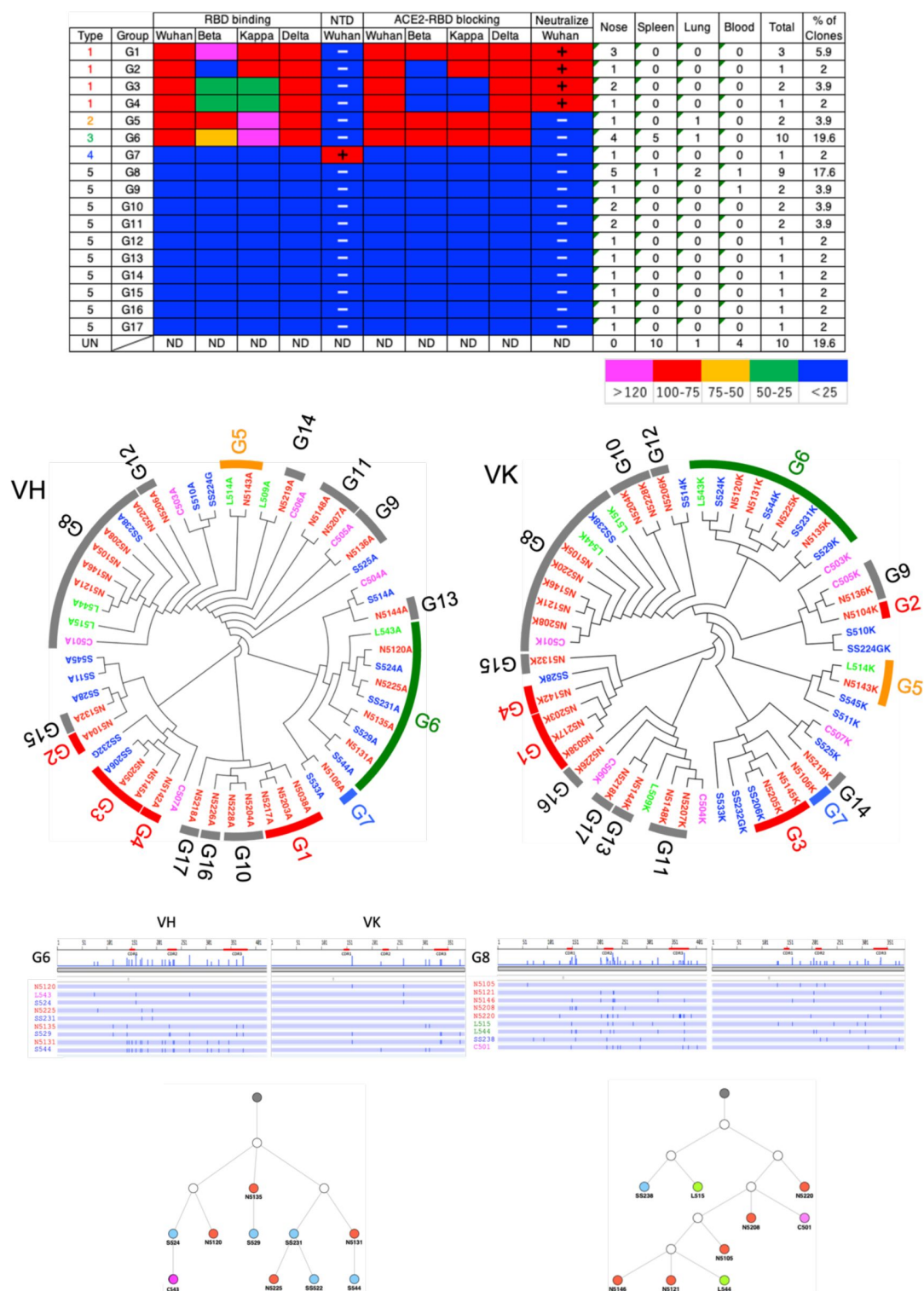


Fig. 3.

Characterization of S1-specific monoclonal antibodies obtained from No. 2 mouse.

(A) A total of 51 S1-reactive antibodies were analyzed for their properties as in Fig 2A. UN, antibody type not determined. ND, antibody activity not determined.

(B) Maximum-likelihood phylogenetic tree of the VH and VL chains of the S1-specific antibodies. Different colored fonts indicate antibodies obtained from the nose (red), spleen (blue), lung (green) and blood (magenta). Antibody groups are indicated by bands on the outer ring. The color of the band indicates antibody types: Type 1 (red), Type 2 (orange), Type 3 (green), Type 4 (blue) and Type 5 (gray). The prefixes N, S and L in the antibody clone numbers refer to antibodies derived from the nose, spleen and lung, respectively. The suffixes A, G and K in the antibody clone numbers refer to alpha, gamma and kappa chain, respectively.

(C) Nucleotide sequence arraignment of VH and VL genes in the G6 and G8 antibodies from No. 2 mouse. The nucleotide changes relative to N5120A and N5105A are depicted as vertical bars across the horizontal lines. Different colored fonts indicate antibodies derived from nose (red), spleen (blue), lung (green) and blood (magenta). Antibody phylogenetic trees based on VH/VK paired sequences are depicted. Gray circles represent the hypothetical germline configuration. White circles represent hypothetical ancestors. Colors indicate nasal (red), splenic (blue), lung (green) and blood (magenta) antibodies.

Multimerization of M-IgA enhances antigen-binding activity

To examine how IgA multimerization affects antigen-binding activity, four representative clones (N5203, N142, N114 and N217) were selected from each type of antibody, and S-IgAs were expressed by cotransfecting alpha heavy chain, kappa light chain, J-chain and secretory component into CHO-S cells. Analysis of the purified S-IgAs by polyacrylamide gel electrophoresis (SDS-PAGE) revealed a band corresponding to the alpha heavy chain, kappa light chain and secretory component. Native PAGE analysis revealed bands corresponding to the dimer (~400 kDa), trimer (~550 kDa) and tetramer (~750 kDa) at a molar ratio of 5:1:3 (**Fig. 4A**). Then, we compared the binding kinetics of each pair of M-IgA/S-IgA by surface plasmon resonance (SPR) with immobilized Spik_{Wuhan}, Spik_{Delta} or Spik_{Omicron}. As shown in **Fig. 4B**, N5203 M-IgA, which has moderate binding affinity to Spik_{Wuhan} (apparent equilibrium constant, $K_D=5.2E-9$), acquired dramatic enhancement of the binding activity to Spik_{Wuhan} after multimerization ($K_D=1.3E-13$). However, multimerization did not enhance the binding activity to Spik_{Delta} ($K_D=2.1E-9$). N142 M-IgA, which has high affinity for Spik_{Wuhan} ($K_D=4.5E-11$) and Spik_{Delta} ($K_D=7.8E-11$), did not show enhanced binding activity to them after multimerization. Almost the same phenomenon was found for N114 M-IgA, which has high affinity for Spik_{Wuhan} ($K_D=6.8E-11$) and Spik_{Delta} ($K_D=9.7E-11$). N142 M-IgA, which has the lowest affinity to Spik_{Wuhan} ($K_D=2.1E-8$), also acquired dramatic enhancement of the binding activity to Spik_{Wuhan} after multimerization ($K_D=6.4E-11$) but not to Spik_{Delta}. All antibodies showed little or marginal levels of binding to Spik_{Omicron}. These results suggest that the degree of avidity of S-IgAs depends on the affinity of the parent monomeric antibody: antibodies with low or intermediate affinity in the monomeric state (N5203 and N217) can increase their avidity by multimerization but not antibodies with high affinity in the monomeric state (N142 and N114). These results are consistent with recent work by Saito, S. et al., who examined the function of multimerized IgA against influenza viruses ([Saito et al., 2019](#)).

Multimerization facilitates stronger neutralization activity in nonneutralizing M-IgA

We next examined whether multimerization of IgAs influences the RBD-ACE2 interaction by competitive ELISA (**Fig. 5A**). Although the multimerization of N5203 M-IgA led to the dramatic enhancement of Spik_{Wuhan} binding, it only increased RBD_{Wuhan}-ACE2 blocking by 3.3-fold, and there were no differences in RBD_{Delta}-ACE2 blocking between M-IgA and S-IgA. In N142 M-IgA, multimerization slightly increased RBD_{Wuhan}-ACE2 blocking but not RBD_{Delta}-ACE2 blocking.

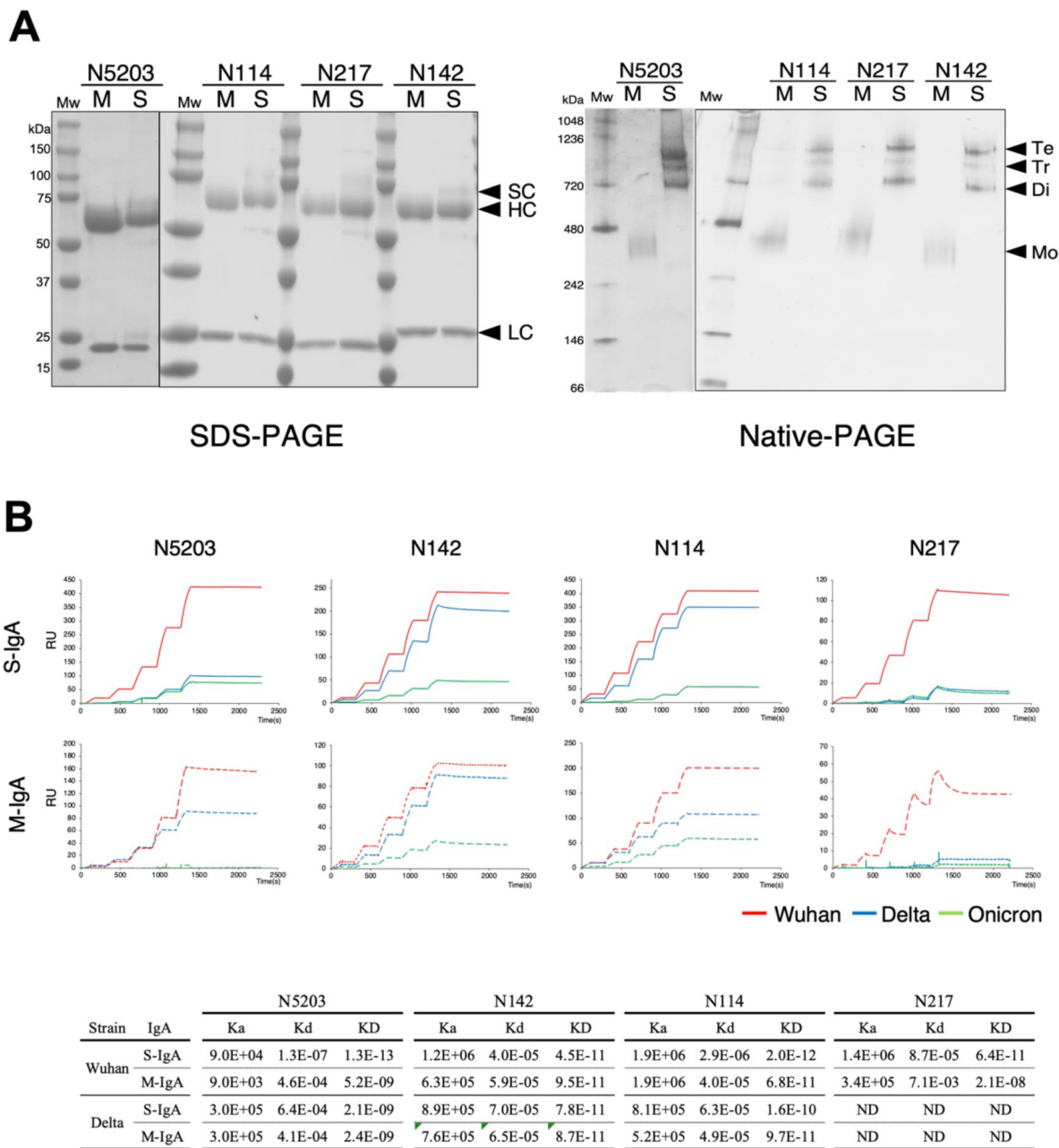


Fig. 4.

Characterization of S1-specific monoclonal antibodies obtained from No. 2 mouse.

(A) A total of 51 S1-reactive antibodies were analyzed for their properties as in **Fig 2A**. UN, antibody type not determined. ND, antibody activity not determined.

(B) Maximum-likelihood phylogenetic tree of the VH and VL chains of the S1-specific antibodies. Different colored fonts indicate antibodies obtained from the nose (red), spleen (blue), lung (green) and blood (magenta). Antibody groups are indicated by bands on the outer ring. The color of the band indicates antibody types: Type 1 (red), Type 2 (orange), Type 3 (green), Type 4 (blue) and Type 5 (gray). The prefixes N, S and L in the antibody clone numbers refer to antibodies derived from the nose, spleen and lung, respectively. The suffixes A, G and K in the antibody clone numbers refer to alpha, gamma and kappa chain, respectively.

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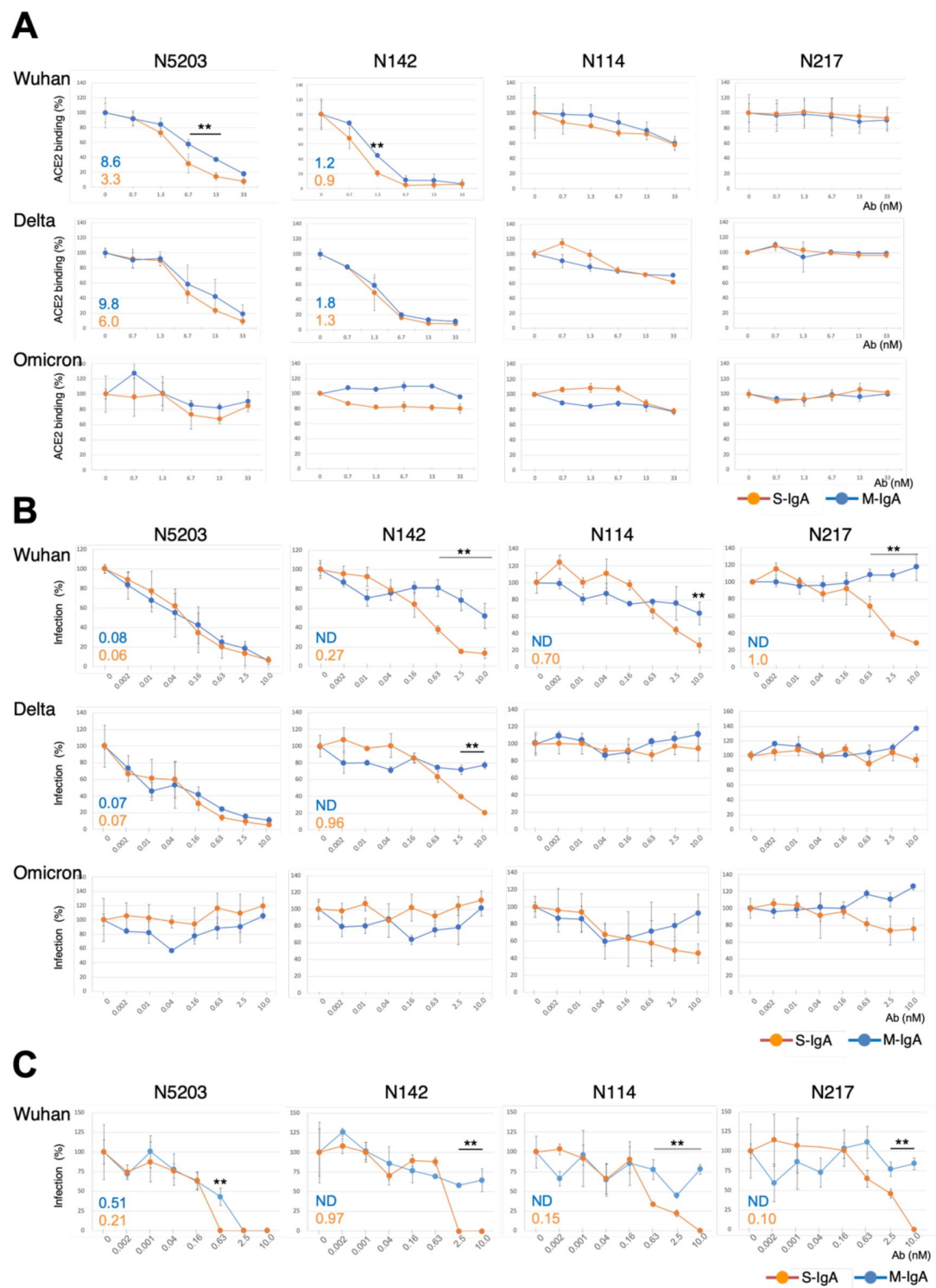


Fig. 5

Multimerization facilitates the neutralization activity of nonneutralizing M-IgAs.

(A) Graphs of the competitive ELISA results showing the binding of biotinylated ACE2 to the immobilized Wuhan, Delta or Omicron RBD in the presence of antibodies. The results are expressed as the mean $\pm$ SD of three technical replicates. The IC₅₀ values of the indicated antibodies that inhibit the RBD-ACE2 interaction are shown in the diagrams.

(B) Comparison of neutralization activity between M-IgA and S-IgA against SARS-CoV-2 pseudotyped viruses. Neutralization curves of the indicated antibody against pseudotyped viruses bearing spike proteins of Wuhan, Delta or Omicron are shown. Pseudotyped viruses preincubated with antibodies at the indicated concentrations were used to infect VeroE6 cells, and luciferase activities in cell lysates were measured at 20 h post transduction to calculate infection (%) relative to nonantibody-treated controls. The results are expressed as the mean $\pm$ SD of three technical replicates. The NT₅₀ values of the indicated antibodies are shown in the diagrams. Antibodies that did not reach >70% inhibition at the highest concentration tested were listed as data not determined (ND).

(C) Comparison of neutralization potential between M-IgA and S-IgA against authentic SARS-CoV-2 BA.1. The neutralizing potential of the antibody was determined using an RT-PCR-based SARS-CoV-2 neutralization assay. VeroE6 cells preincubated with authentic SARS-CoV-2 BA.1 virus were incubated with the indicated antibodies at various concentrations. The virus in the cell culture medium was measured at 48 h post transduction to calculate infection (%) relative to nonantibody-treated controls. The results are expressed as the mean $\pm$ SD of three technical replicates. The NT₅₀ values of the indicated antibodies are shown in the diagrams. Antibodies that did not reach >50% inhibition at the highest concentration tested are listed as ND. ** $p < 0.01$.

Multimerization of N114 and N217 M-IgA did not enhance RBD_{Wuhan}-ACE2 and RBD_{Delta}-ACE2 blocking, as in the case of their monomeric forms. None of the antibodies showed RBD_{Omicron}-ACE2 blocking even at the highest antibody concentrations. To analyze whether multimerization affects the functionality of M-IgAs, we tested all M-IgA/S-IgA pairs for their ability to neutralize pseudotyped lentiviruses bearing the Wuhan, Delta or Omicron spike protein. Both monomeric and multimeric forms of N5203 showed strong neutralizing activity against the Wuhan and Delta pseudotyped lentiviruses, but the concentrations required to achieve the 50% neutralizing titer (NT₅₀) were almost the same for both. Although both forms of N142 displayed RBD_{Wuhan}-ACE2 and RBD_{Delta}-ACE2 blocking, only the multimeric form showed neutralization activity against the Wuhan and Delta pseudotyped lentiviruses. Notably, N114 M-IgA and N217 M-IgA, which did not have RBD_{Wuhan}-ACE2 blocking activity, displayed neutralizing activity against the Wuhan but not the Delta pseudotyped lentivirus after multimerization. None of the antibodies exhibited neutralization activity against the Omicron pseudotyped lentivirus. We further tested the neutralization activity of these antibodies by using an authentic SARS-CoV-2 Wuhan strain. N5203 S-IgA showed twofold higher neutralization activity than its monomeric counterpart. As predicted from the results of the pseudotyped lentivirus assay, multimerization of N142, N114 and N217 M-IgA induced neutralization activity, while their monomeric counterparts failed to neutralize live virus at the highest antibody concentration. These results suggest that the neutralizing activity of S-IgAs is not solely attributed to the spike protein affinity and the RBD-ACE2 blocking activity of M-IgAs, and that the valence and binding mode between the epitope and paratope affect their function.

Discussion

In this paper, we developed 99 antigen-specific monoclonal IgAs from the nasal mucosa of mice intranasally immunized with SpikWuhan and demonstrated that immunization induces functionally diverse antibodies in nasal mucosa. We also cloned 114 antigen-specific antibodies from plasma cells localized in the spleen, lung and blood of the immunized mice, which revealed the presence of many nose-related sequences. Analysis of the reactivity and functionality of the antibodies recognizing different epitopes revealed that multimerization facilitates the neutralizing activity of nonneutralizing M-IgAs. The different effects of multimerization suggest that each S-IgA has a different mechanism of action with respect to the suppression of SARS-CoV-2 infection, and the affinity, valence, and binding mode of the epitope and paratope may contribute to this variation.

It has been shown that the mucosal route of immunization elicits immune responses at local sites, as well as systemic immune responses. (Lapiente *et al.*, 2021). Our analysis demonstrated that the spleen possessed a larger population of antigen-specific plasma cells that expressed nose-related IgAs and IgGs. These results suggest that the B cells activated in the nasal immune system give rise to plasma cells that reside in the nasal mucosa and produce S-IgA, while the B cells also migrate to the spleen and produce M-IgA. The role of the spleen in the induction of nose-related antibodies by intranasal immunization is yet to be determined. The antigen specific plasma cells in spleen may respond in cases of systemic infection by supplying M-IgAs and IgGs into the bloodstream or protecting the body against subsequent infection (Afkhami *et al.*, 2022) (Sheikh-Mohamed *et al.*, 2022). Although bone marrow and the intestines are reservoirs for IgA-producing B cells, we could not detect antigen-specific plasma cells in these tissues. The molecular mechanisms regulating nose-related B-cell migration into the spleen but not into bone marrow and Peyer's patches remain to be determined.

To date, SARS-CoV-2 research has mainly focused on *anti-RBD* neutralizing antibodies, and the role of nonneutralizing antibodies has not been adequately analyzed. In this study, we demonstrated that nonneutralizing M-IgAs, which comprise 70% of the nasal IgA repertoire, showed strong neutralizing activity when expressed as S-IgAs. Our findings suggest that *factors other than the ability to block the ACE2 binding site of RBDs are involved in the neutralization activity of S-IgAs* (Nicasio *et al.*, 2012) (Tan *et al.*, 2016). Recent reports show that antibodies that induce inter- or intraspine crosslinking can inhibit viral binding or shedding by the host cell through steric hindrance or cause conformational changes in the spike proteins (Galimidi *et al.*, 2015) (Jackson *et al.*, 2022) (Klein and Bjorkman, 2010). Such inter- and intraspine cross-linking may be limited in M-IgAs if the spike protein density on the virus surface is low or if the epitopes are unfavorably located. We hypothesize that S-IgA could exert the above effects through its multivalent arms, a mode of action that is hard to achieve with M-IgA, and factors such as valency, epitope selection, antibody binding angle, and the bulkiness of the Fc are involved in this process. (Okuya *et al.*, 2020a) (Callegari *et al.*, 2022).

There are limitations of our study, including the number of antigen-specific plasma cells isolated from the nasal mucosa of immunized mice was insufficient for a comprehensive analysis of systemic and mucosal immune responses. In this study, antibodies that did not share V(D)-J with nasal clones were found in the spleen, lungs and blood of immunized mice. This finding was probably due to the limited number of clones that was isolated from the nose. However, we cannot exclude the possibility that these clones were differentiated from local naïve B cells.

In conclusion, our study highlights the key role of S-IgA in the protective effect of mucosal immunity, which may be useful for better understanding how intranasal vaccines can help protect against SARS-CoV-2 infection.

Materials and methods Study approval

All experiments were performed in accordance with relevant guidelines and regulations. Animal experimental protocols were approved by the Committee on Animal Experimentation at the University of Toyama and conducted using project license A2017eng-1.

Materials

The materials used in this study can be found in the Supplementary Ma. VeroE6/TMPRSS2 cells were purchased from the Japanese Collection of Research Bioresources (JCRB) Cell Bank (the National Institute of Biomedical Innovation, Health, and Nutrition, Osaka, Japan) (JCRB1819). Female ICR mice were purchased from Japan SLC, Inc. (Tokyo, Japan).

Immunization

Anesthetized mice were intranasally immunized with 5 μg of SARS-CoV-2 S protein trimer and 0.1 μg of cholera toxin in 10 μl of PBS by using a pipette to deliver the fluid dropwise into each nostril a total of 3 times at 3-week intervals. One week after the final immunization, mice were sacrificed by CO_2 inhalation, and blood samples were collected from the inferior vena cava. To avoid contamination with circulating blood lymphocytes, the mice were perfused with 2 ml of PBS via the heart. Then, the lungs, the spleen, bone marrow and Peyer's patches were dissected, and single-cell suspensions were obtained by mincing the tissue using a 100- μm nylon mesh. The mice were decapitated at the larynx. After the removal of the facial skin from the head, the nose was separated from the rest of the head. A pipette tip was inserted through the pharyngeal opening into the choana, and then two consecutive volumes of 250 μl of PBS were gently perfused, and the nostril fluid was collected in a tube. The nasal lavage fluids were centrifuged, and supernatants were stored at -80°C until assayed. The isolated nose was cut into two along the midline, the layer of epithelium was mechanically removed from the nasal septum by gently rubbing the sample with a needle under a stereoscopic microscope, and the tissues were then mechanically shredded. Single-cell suspensions containing lymphocytes were isolated by using Lympholyte-M (Cedar Lane, Ontario, Canada).

Elisa

ELISA to evaluate antibody binding to the S-trimer, RBDs or NTD was performed by coating Nunc MaxiSorp flat-bottom high-binding 96-half-well plates (ThermoFisher Scientific, MS, USA) with 50 μl per well of a 0.1 $\mu\text{g ml}^{-1}$ protein solution in PBS overnight at 4°C . Plates were washed 3 times with PBS and incubated with 170 μl per well Blocking One solution (Nakarai, Tokyo, Japan) for 1 h at room temperature. Serially diluted antibody, serum or nasal lavage fluid was added to PBST (1 $\times$ PBS with 0.1% Tween-20) and incubated for 1 h at room temperature. The plates were washed three times with PBST and then incubated with anti-mouse IgG or IgA secondary antibody conjugated to horseradish peroxidase (HRP) (Abcam, Cambridge, UK) in PBST for 1 h at room temperature. After washing with PBST 3 times, the plates were developed by addition of the SureBlue/TMB peroxidase substrate and stop solution (KPL, MS, USA), and absorbance was measured at 450 nm with an ELISA microplate reader. Serum and nasal fluid antibody titers were also determined ELISA. 96-well plates coated with 1 μg of anti-mouse IgA or IgG were blocked with Blocking One solution. Samples were diluted with PBST and incubated for 1 h at room temperature. After washing, 100 μL alkaline phosphatase-conjugated anti-mouse IgA or anti-mouse IgG was added to each well and incubated for 1 h at room temperature. Plates were washed and developed with BluePhos Microwell Substrate Kit (KPL, MS, USA). Mouse IgA or IgG was used as a reference to construct a standard curve for quantifying antibodies.

ACE2 blocking assay

Nunc Maxisorp plates were coated with SARS-CoV-2 RBD at 50 ng per well and incubated overnight at 4°C . After blocking with Blocking One solution for 1 h at room temperature, serially diluted antibody mixed with 5 ng of biotinylated ACE2 in 100 μL of PBST was transferred to the plate in triplicate. After incubation for 1 h at room temperature, the assay plate was washed with PBST three times, and 100 μL of streptavidin-HRP (Abcam, Cambridge, UK) diluted 1:5000 in PBST was transferred to each well and incubated for 30 min. After three washes, the plate was developed with streptavidin-HRP and SureBlue/TMB peroxidase substrate.

Plasmid construction

The DNA fragments encoding the human IgA2 constant region, human J-chain and extracellular domain of human pIgR were amplified by PCR. The IgA2 constant region was replaced with the IgG1 constant region of pJON-mIgG or pETmIgA to make pJON-hIgA and pET-hIgA, respectively

(Kurosawa *et al.*, 2012 [DOI](#)). The human J chain was replaced with the DsRed2 gene of pDsRedN1 (Takara Bio, Shiga, Japan). The pIgR gene was inserted into the pEF-Myc-His vector (ThermoFisher Scientific, MS, USA).

Isolation of antigen-specific plasma cells

Isolation of antigen-specific plasma cells was performed as described previously with slight modifications (Kurosawa *et al.*, 2012). Nasal lymphoid cells or splenocytes were stained with PE-labeled anti-mouse CD138, DyLight-650-labeled S1 and DyLight488-labeled anti-mouse IgA or anti-mouse IgG at 4°C for 30 min with gentle agitation. After washing with PBS, the cells were suspended in PBS containing ER-tracker and subsequently analyzed by FACS. The forward-versus-side-scatter (FSC vs. SSC) lymphocyte gate (R1) was applied to exclude dead cells. The plasma cells (IgGA+, CD138+, ERhigh, R2 gate) were further subdivided into fractions according to their binding of fluorescently labeled SARS-CoV-2 to define antigen-specific plasma cells (IgA+, ERhigh SARS+). Single-cell sorting was performed using a JSAN Cell Sorter that was equipped with an automatic cell deposition unit (JSAN, Kobe, Japan) with DyLight488-labeled antibodies against IgA monitored in the FL-1 channel, PE-labeled CD138 in the FL-2 channel, ER-tracker in the FL-7 channel and DyLight650-labeled SARS-COV-2 in the FL-6 channel.

Monoclonal antibody generation

Molecular cloning of VH and VL genes from single cells was performed by 5'-RACE PCR as previously described. For the first antibody screening, the PCR-amplified VHa and VL genes were joined to pJON-hIgA and pJON-mIgK to make full-length immunoglobulin alpha heavy and kappa light chain genes by TS-jPCR, respectively (Yoshioka *et al.*, 2011). M-IgA was expressed by transfecting a pair of alpha heavy and kappa light chain genes into FreeStyle CHO-S cells that were grown in a 24-well plate according to the manufacturer's protocol (CHOgro High Yield Expression System, Takara Bio, Shiga, Japan). For large-scale antibody production, the PCR-amplified alpha heavy and kappa light chain genes were inserted into pET-hIgA and pET-mIgK by TS-HR, respectively (Kurosawa *et al.*, 2011).

M-IgA was expressed by cotransfecting plasmids encoding the IgA gene (5 µg) and IgK gene (5 µg) into FreeStyle™ CHO-S cells (2.0×10^5 cells/10 ml). S-IgA was expressed by cotransfecting plasmids encoding the IgA gene (5 µg), IgK gene (5 µg), J-chain (1 µg) and pIgR gene (1 µg) into FreeStyle™ CHO-S cells (2.0×10^5 cells/10 ml). M-IgA was purified by using Peptide M Agarose (ThermoFisher Scientific). S-IgA was purified by two-step chromatography using the Capturem His-Taged Purification Kit (Takara Bio, Shiga, Japan) followed by size exclusion chromatography (Cytiva, Acta go, MS, USA). The purified antibodies were analyzed by SDS-PAGE and native polyacrylamide gel electrophoresis on NuPAGE 4–12% Bis-Tris gels (Thermo Fisher Scientific MS, USA).

Antibody binding kinetics by SPR

The binding kinetics and affinity of monoclonal antibodies to the SARS-CoV-2 spike trimer were analyzed by SPR (Biacore T100, GE Healthcare). Specifically, a biotinylated spike trimer was covalently immobilized to an SA Sensor Chip for a final RU of approximately 200 and interacted with M-IgA or S-IgA at various concentrations (0.3, 1.0, 3.0, 9.0 and 27 nM of each antibody). SPR assays were run at a flow rate of 30 µl/min in HEPES buffer. The dissociation phase was monitored for 5 minutes. The sensograms were fit in a two-component model with BIA Evaluation software (GE Healthcare).

Neutralization activity of monoclonal antibodies against pseudotyped SARS-CoV-2 virus

VeroE6/TMPRSS2 cells were incubated with serially diluted antibodies and pseudotyped virus possessing the spike protein of the Wuhan, Delta, Omicron strains or vesicular stomatitis virus (VSV) and cultured for 48 h at 37°C. After exposure to the virus–antibody mixture, the infectivity of the pseudotyped viruses was determined by measuring the luciferase activities using a PicaGene Luminescence Kit (Fujifilm Wako, Osaka, Japan) with a GloMax Navigator Microplate Luminometer (Promega, WI, USA).

Neutralization assay using an authentic virus strain

The neutralizing activity of monoclonal antibodies against an authentic Wuhan SARS-CoV-2 strain was determined by a neutralization test in a biosafety level 3 laboratory at the Toyama Institute of Health as previously described (Ozawa *et al.*, 2022). VeroE6/TMPRSS2 cells plated at 2×10^4 cells in each well of 96-well plates were infected with the Wuhan SARS-CoV-2 strain at a multiplicity of infection of 0.001 per cell in the presence of serially 2-fold diluted monoclonal antibodies for 1 h. After discarding the culture supernatants, cells were cultured for 24 h with DMEM containing 10% FBS in the presence of the indicated concentration of monoclonal antibodies. The viral infectious dose was determined by the level of viral genomic RNA in the culture supernatant, which was measured using a real-time PCR assay with a SARS-CoV-2 direct detection RT-qPCR kit (Takara Bio, Siga, Japan). The IC₅₀ was calculated by IC50 Calculator (<https://www.aatbio.com/tools/ic50-calculator>) and represented the neutralization titer.

Gene family and phylogenetic analysis of monoclonal antibodies

The antibody sequences were annotated against the IMGT mouse heavy and light chain gene database using NCBI IgBlast to determine IGHV, IGHD, IGLV, IGHJ, IGLV and IGLJ gene annotations. Antibody clones were assigned to clonal groups using Sequencher software. The heavy and light chain variable gene arrangement and phylogenetic analyses were performed using MAFFT, a multiple alignment program, in the GENETYX sequence analysis package (<https://www.genetyx.co.jp>). For somatic hypermutation, IGHV and IGLV sequences were aligned against representative clones of each group. Antibody clones consisting of pairs of heavy and light chain variable genes (signal sequence to FW4) were used to generate an antibody phylogenetic tree. Full-length germline sequences were reconstructed, with nucleotide additions/deletions in the junction between V-(D)-J adjusted to match the sequence of each antibody group. Within these groups, if the combined sequence of the heavy and light chains differed by more than five bases from each other, they were defined as separate clones. The sequence alignment tool Clustal Omega (<https://www.ebi.ac.uk/Tools/msa/clustalo/>) was used to identify deletions and insertions and to align the length of the sequence. The nucleotide lengths were aligned by adding “-” to the missing bases. The data file required by Alakazam was created in RStudio (version 2022.12.0+353), and a phylogenetic tree was created by Alakazam (<https://www.rdocumentation.org/packages/alakazam/versions/1.2.1> and <https://alakazam.readthedocs.io/en/stable/>). The R script was as follows: <https://www.rdocumentation.org/packages/alakazam/versions/1.2.1/topics/buil> dPhylypLineage

Statistical analysis

All statistical analyses were performed using JMP statistical software (JMP Statistical Discovery, NC, USA). Unpaired Student’s t tests were used to analyze each dataset. The threshold for statistical significance was set at $p < 0.01$ (**).

Data availability

The nucleotide sequence data were submitted to the DDBJ/EMBL/GenBank databases (Accession No. LC761389 ~ LC761570). Other relevant data are available from the corresponding authors upon request.

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Reviewer #1 (Public Review):

Despite evidence suggesting the benefits of neutralizing mucosa-derived IgA in the upper airway in protection against the SARS-CoV-2 virus, all currently approved vaccines are administered intramuscularly, which mainly induces systemic IgG. Waki et al. aimed to characterize the benefits of intranasal vaccination at the molecular level by isolating B cell clones from nasal tissue. The authors found that Spike-specific plasma cells isolated from the spleen of vaccinated mice showed significant clonal overlap with Spike-specific plasma cells isolated from nasal tissue. Interestingly, they could not detect any spike-specific plasma cells in the bone marrow or Peyer's patches, indicating that these nose-derived cells did not necessarily home to and reside in these locations, although the Peyer's patch is not a typical plasma cell niche - rather the lamina propria of the gut would have been a better place to look. Furthermore, they found that multimerization improves the antibody/antigen binding when the antibody is of low or intermediate affinity, but that high-affinity monomeric antibodies do not benefit from multimerization. Lastly, the authors used a competitive ELISA assay to show that multimerization could improve the neutralizing capacity of these antibodies.

The strength of this paper is the cloning of multiple IgA from the nasal mucosae (n=99) and the periphery (n=114) post-SARS-CoV-2 i.n. vaccination to examine the clonal relationship of this IgA with other sites, including the spleen. This analysis provides novel insights into the nature of the mucosal antibody response at the site where the host would encounter the virus, and whether this IgA response disseminates to other tissues.

There were also some weaknesses:

1. The finding that multimerization improves binding and neutralization is not surprising as this was observed before by Wang and Nussenzweig for anti-SARS-CoV-2 IgA (authors should cite Enhanced SARS-CoV-2 neutralization by dimeric IgA. Wang et al, Sci. Transl. Med 2021, 13:3abf1555). In addition, as far as I can tell we cannot ascertain the purity of fractions from the size exclusion chromatography thus I wasn't sure whether the input material used in Fig. 4 was a mixed population of dimer/trimer/tetramer?
2. The flow cytometric assessment of the IgA+ clones from the nasal mucosae was difficult to interpret (Fig. 1B). It was hard for me to tell what they were gating on and subsequently analysing without an IgA-negative population for reference.

3. While the i.n. study itself is large and challenging, it would have been interesting to compare an i.m. route and examine the breadth of SARS-CoV-2 variant S1 binding for IgGs as in Fig. 2A. Are the IgA responses derived from the mucosae of greater breadth than systemic IgG responses? Alternatively, and easier, authors could do some comparisons with well-characterized IgG mAb for affinity and cross-reactivity as a benchmark to compare with the IgAs they looked at.

Overall the authors did a good job of looking at a large range of systemic vs mucosal S1-specific antibodies in the context of an intra-nasal vaccination and this provides additional evidence for the utility of mucosal vaccination approaches for reducing person-to-person transmission.

Reviewer #2 (Public Review):

Summary:

This research demonstrates the breadth of IgA response as determined by isolating individual antigen-specific B cells and generating mAbs in mice following intranasal immunization of mice with SARS-CoV2 Spike protein. The findings show that some IgA mAb can neutralize the virus, but many do not. Notable immunization with Wuhan S protein generates a weak response to the omicron variant.

Strengths:

Detailed analysis characterizing individual B cells with the generation of mAbs demonstrates the response's breadth and diversity of IgA responses and the ability to generate systemic immune responses.

Weaknesses:

The data presentation needs clarity, and results show mAb ability to inhibit SARS-CoV2 in vitro. How IgA functions in vivo is uncertain.